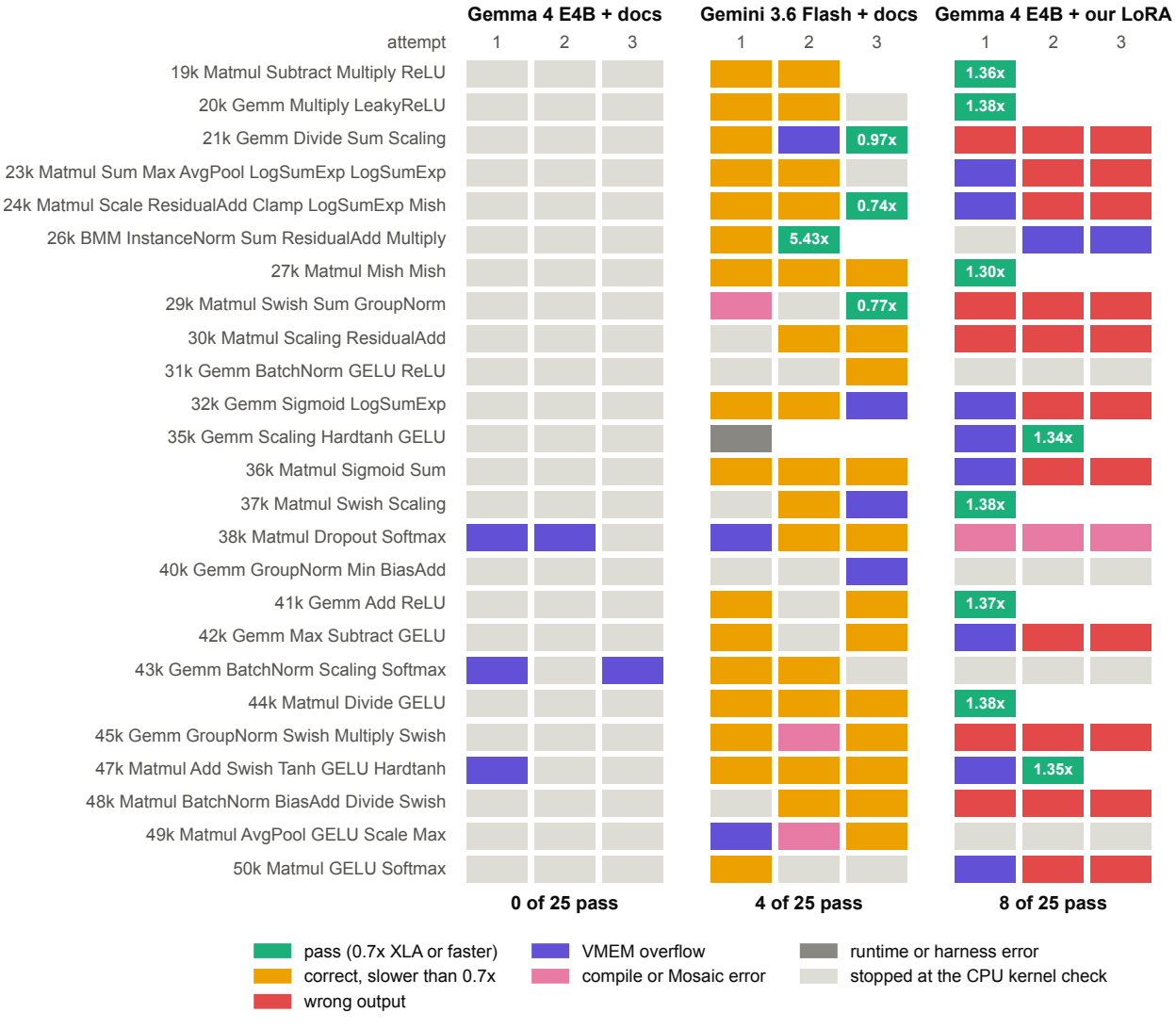
Gemma 4 E4B + docs
Gemini 3.6 Flash + docs
Gemma 4 E4B + our LoRA
attempt
1 2 3
1 2 3
1 2 3
19k Matmul Subtract Multiply ReLU
20k Gemm Multiply LeakyReLU
21k Gemm Divide Sum Scaling
23k Matmul Sum Max AvgPool LogSumExp LogSumExp
24k Matmul Scale ResidualAdd Clamp LogSumExp Mish
26k BMM InstanceNorm Sum ResidualAdd Multiply
27k Matmul Mish Mish
29k Matmul Swish Sum GroupNorm
30k Matmul Scaling ResidualAdd
31k Gemm BatchNorm GELU ReLU
32k Gemm Sigmoid LogSumExp
35k Gemm Scaling Hardtanh GELU
36k Matmul Sigmoid Sum
37k Matmul Swish Scaling
38k Matmul Dropout Softmax
40k Gemm GroupNorm Min BiasAdd
41k Gemm Add ReLU
42k Gemm Max Subtract GELU
43k Gemm BatchNorm Scaling Softmax
44k Matmul Divide GELU
45k Gemm GroupNorm Swish Multiply Swish
47k Matmul Add Swish Tanh GELU Hardtanh
48k Matmul BatchNorm BiasAdd Divide Swish
49k Matmul AvgPool GELU Scale Max
50k Matmul GELU Softmax
0.97x
0.74x
5.43x
0.77x
1.36x
1.38x
1.30x
1.34x
1.38x
1.37x
1.38x
1.35x
0 of 25 pass
4 of 25 pass
8 of 25 pass
pass (0.7x XLA or faster)
correct, slower than 0.7x
wrong output
VMEM overflow
compile or Mosaic error
runtime or harness error
stopped at the CPU kernel check